Week 7 - Practical test prepare

Practical 1

Practical 1- slide

Basic Shell Commands (continued)

Command	Description
`clear`	Clears the terminal screen
`reset`	Resets your terminal to use default terminal settings
`who`	Displays currently logged-in users
`w`	Displays currently logged-in users and their tasks
`whoami`	Displays your login name
`id`	Displays the numbers associated with your user account name and group names; these are commonly referred to as User IDs (UIDs) and Group IDs (GIDs)
`date`	Displays the current date and time
`cal`	Displays the calendar for the current month
`uname -a`	Displays system information
`ls`	Lists files
`exit`	Exits out of your current shell

Basic Shell Commands - Maintain the system time

- To view and set date and time

Command	Function
date	To list the date and time of the server.
date -s "*datestr*"	To set time and date to the value specified. Eg. "11/20/2026 12:48:00"

**set time requires superuser privilege*

Continue...

- To show calendar

Command	Function
cal	To show calendar of the month.
cal *month year*	To show calendar of the specified month and year.

Practical 1 - practice

3. Fill in the following UNIX commands:

Command	Function	Note
date	To list the date and time of the server.	
date -s "datestr"	To set time and date to the value specified. Eg. "11/20/2025 12:48:00"	
cal month year	To show the calendar of the specified month and year.	
man	To provide in depth information about the requested command.	
command -help	To display a short help file.	
clear	To clear the screen.	
pwd	To display the name of the current working directory.	
sudo password	To allow a user to execute a command as the superuser.	
exit	To leave the current shell session.	

Practical 2

Practical 2 - slide

Continue...

- Directories navigation
- cd command: To change / switch directories.

Command	Function
cd	Change from any directory to the home directory
cd /home	Go to the directory /home (absolute path)
cd ~	Go to /home/*user (the user's home directories)*
cd /	Go to root (NOT "/root")
cd ..	To move one directory level higher
cd ../..	To move two directory levels higher
cd -	To go to the last valid directory

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Listing Files

ls command: List the files in a directory

- ✓ May pass an argument indicating the directory to be listed

Option	Description
(None)	Displays the content of the current directory in several columns
-a	Also displays hidden files (such as .bashrc)
-F	After each name, a character indicates the file type
-l	Long file listings - gives a detailed list of all files.
-t	File are sorted by date of alteration (newest first). Combined with the option –r, the output takes place in reverse order
-R	Output is recursive, including all subdirectories
-u	Sorted by date of the last access

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Create and View Files

Create a New File with "**touch**" command

Command	Description
touch <file>	Create a file (with no contents)
touch <file1> <file2>	Create 2 or more file
-a	Changes only the time of the last read access
-m	Changes only the time of the last modification
-r file	Set the time stamp of file instead of the current time e.g. touch –r file2 file4 or touch –reference=file2 file4
-t time	Sets time using [[CC]YY]MMDDhhmm[.ss] instead of current time E.g. touch –t 201712251830.00 file5

View/Displaying Content of Text Files

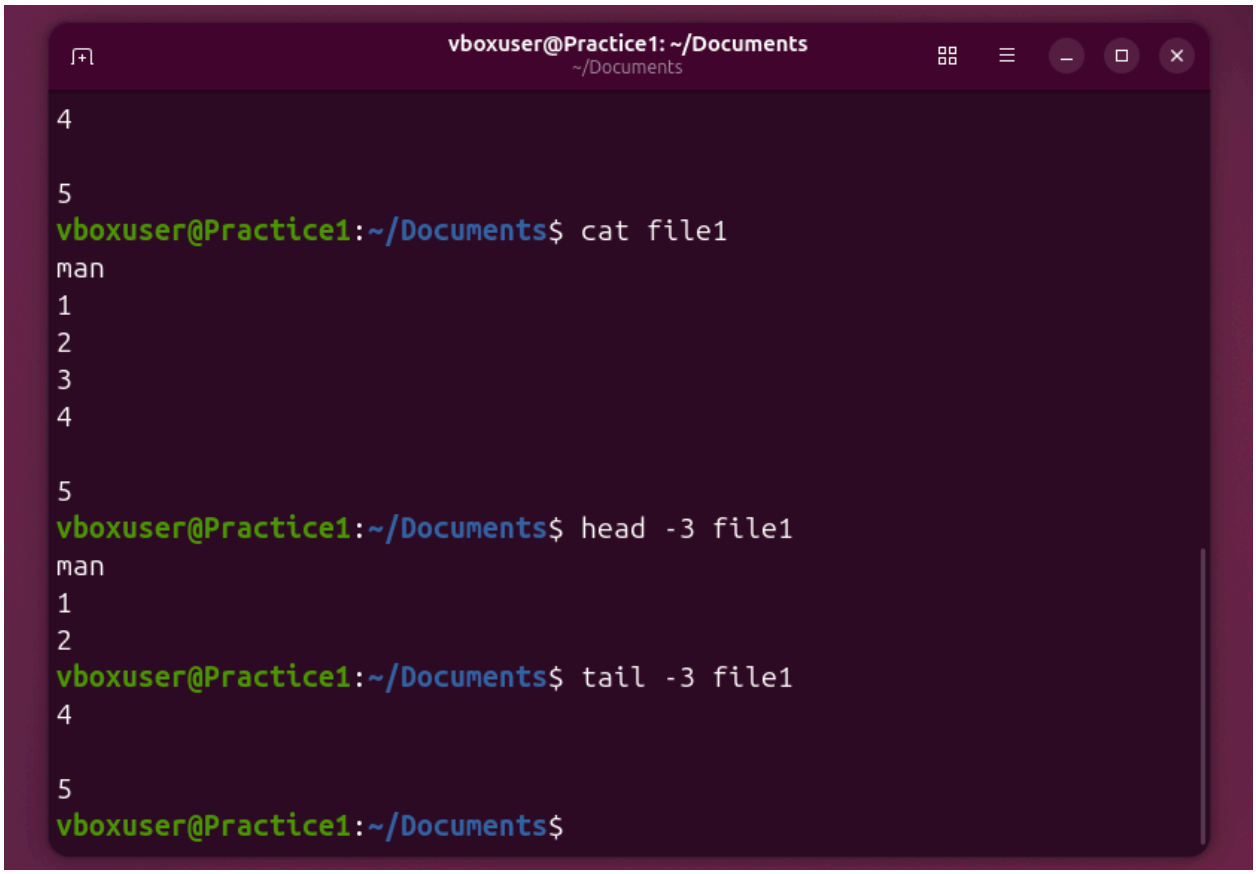
- Large text files can not be viewed using the cat command, because the screen will only fit a portion of the file
- **more** command: displays text files page-by-page
  - Pressing Spacebar displays the next page
  - Pressing Enter displays the next line
- **less** command: same as more command but with more functions, e.g. can also use cursor to scroll. Besides, even compressed files (such as .gz and .bz2) can be displayed

Create and View Files

View a File with "**head**" and "**tail**" command

- Used to view the first or last lines of a file
- By default, they show ten lines
- head -20 or head –n 20 displays the first twenty lines
- tail -5 or tail –n 5 displays the last five lines
- tail -f displays a continuously updated view of the last lines of a file
- To exit tail –f, press Ctrl+c

Command	Description
head -<no of lines> <filename> tail -<no of lines> <filename>	View the file contents



Move Files and Directories

- "mv" command
- mv source destination

Command	Description
mv <filename> <directory>	Move the file / Rename of a file →**mv** renames file/directory *SOURCE* to *DEST*, or moves the *SOURCE* file (or files) into DEST directories or moves directory_A into *DIRECTORY_B*.
Option	**Description**
-i	Ask for confirmation before moving or rename a file
-u	Only moves files that are newer that the target files or when the destination file is missing

Create and Delete Files and Directories

Command	Description
rm <filename>	Remove a file
mkdir <directory name>	Make a directory
mkdir <directory name> <directory name2>	Make 2 or more directory
mkdir **–p** <dir1/dir1.1/dir1.1.1>	Make sub subdirectories
rmdir	rm a directory

Option for rm command	Description
-i	Asks for confirmation before deleting
-r	Allows full directories to be deleted (rm –r <directory name>

Find Files on Linux

- To search for files on the command line, you can use the **find** command.
- The syntax for the find command is *find path criterion action*.
- **Path** → the section of the file system to search (the specified directory and all its subdirectories)
- **Criterion** → The properties of the file should have (refer to your book)
- **Action** → Options that influence the following conditions or control the search as a whole, such as **–print** and **–exec** command.
- E.g.

```
find ~ -name file1 –print
find ~/Desktop -name myfile –exec cat {} \;
```

Search File Content

grep <word> <filename>	Search for the word / content in the file. grep apple file1 -i: Ignores case -l: Shows only the names of files that contain the search string -r: Search entire directory tree recursively -v: Gives all lines that do not contain the search string -n: Shows the line numbers [0-9]: Any one from the specified range. (e.g. match any one number from 0 to 9. [^abc]: None of the characters. (e.g. “a”, “b”, “c” are not matched.) .: Any single character (e.g. file. Matches file1 but not file10)

Search File Content

egrep <word> <filename> Or **grep –E** <word> <filename>	Accepts extended regular expression egrep “(b|B)all” file* egrep “letter+” : matched 1 or more occurrences of previous character (e.g. matches letter1, letterrr, letter and so on) egrep “letter?” : matched 0 or 1 occurrences of previous character (e.g. matches lette, letter and so on egrep “co{1,2}l” : matched both “col” and “cool”

WC

- **wc,** or "word count," prints a count of newlines, words, and bytes for each input file.
- E.g. wc myfile.txt

→Displays information about the file **myfile.txt**. Output will resemble the following:

5 13 57 myfile.txt

Where **5** is the number of lines, **13** is the number of words, and **57** is the number of characters/bytes.

Using gzip/gunzip/bzip2/bunzip2

```
gzip myfile              # removes myfile; creates myfile.gz
gunzip –l myfile.gz      # shows statistics of compressed file
gunzip myfile.gz         # removes myfile.gz; creates myfile


bzip2 myfile             # removes myfile; creates myfile.bz2
bunzip2 myfile.bz2       # removes myfile.gz; creates myfile
```


Tape Archive - TAR

```
tar –cf  doc.tar     *     # c-create an archive under –f archive file name
tar –tf  docs.tar          # t-list an archive's contents
tar –xf  doc.tar           # x-extract files from archive


tar –czf  doc.tar.gz   *   # z-compress an archive with gzip or decompress
                             an archive with gunzip
tar –xzf doc.tar.gz
tar –xf doc.tar  file1     # only extract file1 from foo.tar archive file
```

Practical 2 practice

AMCS2093 Operating Systems

Practical 2 – Managing Files in Linux System

Section A

Starting from the user's home directory (cd ~), perform the following in sequence.

1. Show the current path.
pwd

2. Create a directory "D2".
mkdir D2

3. Give two commands which can list all the contents at the current directory.
ls -a
dir -a

4. Create a directory "super" inside the directory "D2".
cd D2
mkdir super

5. Inside this "super" directory, create four files "apple.doc", "april.doc", "box.txt" and "ultra.txt" using *touch* command.
cd super
Touch apple.doc april.doc box.txt ultra.txt

6. (a) Show the file(s) starting with “ap”.
ls ap*
(b) Show the file(s) that the length of the filename contains 7 characters.
ls ???????
(c) Show the file(s) ending with “txt” extension.
ls *.txt
(d) Show the file(s) that the third character of the filename is “p”.
ls ???p*

7. Go to root directory.
cd /

8. Search a directory named “super” from the root directory.
find / -name super

9. Search a file named “ultra.txt” from the root directory.
find / -name ultra.txt

10. Go back to the directory “super”. Rename the file “ultra.txt” to “mega.txt”
cd/home/tarumt/super mv ultar.txt mega.txt

11. Copy “super” directory to *home* directory and rename it to “super2” in one command line.
cp -r super /home/super2

12. Confirm whether the “super2” directory is created and copied.
ls/home

13. Remove “super” and all its subdirectories and files.
rm -r super

14. Create a text file “sampleA.txt” using *echo* command inside the directory “D2”.
echo “MAN” > D2/sampleA.txt

15. Create another text file “sampleB.txt” using *cat* command.
cat > sampleB.txt

16. Type the following contents into “sampleB.txt” and save the file. *“When using a graphical user interface, we need another program called a terminal emulator to interact with the shell.”*

17. Open “sampleB.txt” again using *cat* command and add the following sentences. *“It’s likely called simply “terminal”.”*

18. Show the word count of “sampleB.txt”
wc -w sampleB.txt

19. Combine the two files “sampleA.txt” and “sampleB.txt” to a file named “sampleC.txt”.
cat sampleA.txt sampleB.txt > sampleC.txt

20. Show the content of “sampleC.txt” on the terminal.
cat sampleC.txt

21. Archive “sampleA.txt” and “sampleB.txt” to “sampleC.tar”
tar -cf sampleC.tar sampleA.txt sampleB.txt

22. Overwrite “sampleB.txt” by “sampleA.txt” and show the content of both files on the terminal.
cat sampleA.txt > sampleB.txt
cat sampleA.txt sampleB.txt

23. Delete the file “sampleB.txt”.
rm sampleB.txt

24. Go back to *home* directory and create a new directory named “backup”.
cd /home
mkdir backup

25. Backup “sampleA.txt” and “sampleC.txt” to “backup” directory.
cp home/tarumt/D2/*.txt backup

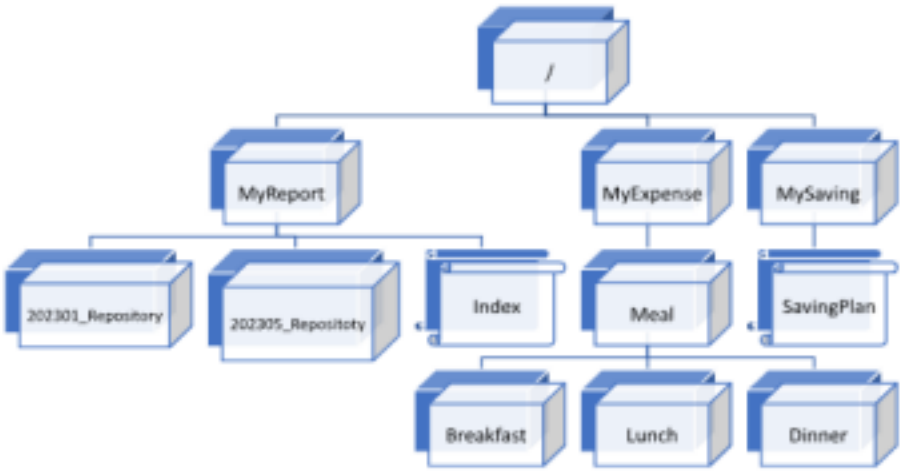
Section B

True/False. Explain the answer if it is false.

		True/False
1.	The structure of the file system in Linux can be depicted in a form of tree. It begins at the root (/), and branches out like the branches of a tree.	True
2.	A relative path starting from the current directory while an absolute path starting from the root of the entire file system tree.	True
3.	When the terminal is opened the starting directory is the root (/) directory.	False
4.	/etc directory in Linux is similar to Control Panel in WINDOWS.	True
5.	Windows separates directories using forward slash (“/”) while LINUX uses back slash (“\”).	
6.	LINUX is non-case sensitive while WINDOWS is.	
7.	Windows uses a volume-based file hierarchy (e.g. C: volume, D: volume) while LINUX uses a unified scheme (e.g. /, /home, /etc).	
8.	.tar is a text file extension in LINUX system.	
9.	LINUX uses color to differentiate the files and folder/directory. Blue is for executable file, Green is for directory.	

10.	Lossless compression preserves all the data after compression while lossy compression drops some information after compression.	

Extra exercises (Optional)

1. 

i. Refer to the directory tree structure as above, if your current working directory is ***Dinner***, write a single line command to reach the ***202305_Repository*** directory using:

Answer (Absolute Path): /MyReport/*202305_Repository*
Answer (Relative Path): ../../../MyReport/*202305_Repository*

ii. Using wildcards token, list only the directories named ***MyReport*** and ***MySaving***.

Answer: ls My??????

iii. Using wildcards token, list all the directories that end with the string "***Repository***" in ***/MyReport***.

Answer: ls /MyReport/*Repository

2. To combine **/etc/passwd** with /**etc/group** and send the output into the file **users_groups.txt Answer: cat /etc/password /etc/group > users_groups.txt**

3. To create a symbolic link of **doc1** file to **softlink** in your home directory
Answer: cp -s doc1 softlink OR ln -s doc1 softlink

4. Display the content of **/var/log/messages** page-by-page.
Answer: more /var/log/messages

5.	Sort the contents of the file called **CourseListing** in *reverse* alphabetical order (Z to A). **Answer: sort -r CourseListing**
6.	Move and rename the **/tmp/file2** file to **~/my_file2** by using a single command.

	Answer: mv /tmp/file2 ~/my_file2
7.	Compress an archive named ***MyArchive.tar.gz*** containing files ***file1*** and ***file2***. **Answer: tar -czf MyArchive.tar.gz file1 file2**
8.	Find all files in the **/home** directory that have the word "AMCS2093" as part of their filename. **Answer: find /home -name "*AMCS2093*"**
9.	List all the directory names that exist under the /var directory **Answer: ls -d /var/***